

LESSON 4-3 • HOMEWORK PACKET • SHEET 1 OF 2

Estimate with Benchmarks

Maryland Standard 6.RP.A.3c • Learning target – I can estimate a percent by comparing an amount to benchmark percents.

REMEMBER

- **Benchmarks:** 10%, 25%, 50%, 75%, 100%.
- Find benchmark amounts of the **whole**. Which one is the part **about** equal to?
- A **double number line** lines up amounts with percents.

WORKED EXAMPLE

A water jug holds 40 cups. There are 29 cups left. About what percent of the jug is left?

Step 1 Benchmarks of 40: $50\% = 20$ and $75\% = 30$.

Step 2 29 is very close to 30.

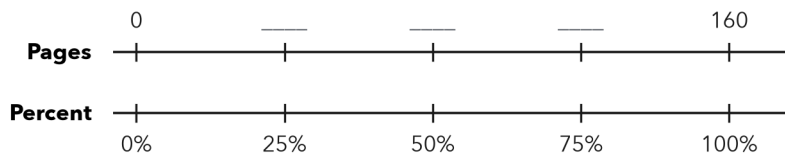
Step 3 So 29 out of 40 is about 75%.

Answer about 75%

PRACTICE

show your work • label units

1. **Reading Goal.** Jada's goal is to read 160 pages. She has read 42 pages. Fill in the double number line. Then estimate the percent of her goal she has read.



About _____ %

2. About what percent? Name the closest benchmark: 10%, 25%, 50%, 75%, or 100%.
- a. 7 out of 60 $\approx$ _____ % b. 38 out of 50 $\approx$ _____ % c. 23 out of 90 $\approx$ _____ %
- d. 19 out of 40 $\approx$ _____ % e. 49 out of 50 $\approx$ _____ % f. 11 out of 45 $\approx$ _____ %
- g. 16 out of 150 $\approx$ _____ % h. 61 out of 80 $\approx$ _____ %
3. **Class Fundraiser.** A class wants to raise \$500. So far they have raised \$96. Use 10% of the goal to estimate the percent they have raised.
- a. 10% of \$500 = _____
- b. About what percent have they raised? _____ %
4. **Bus Seats.** A bus has 48 seats. 35 seats are filled. About what percent of the seats are filled? Name the benchmark you used.
- About _____ %
5. **Compare.** Which is closer to 50%: 26 out of 50, or 26 out of 60? Explain how you know.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Look at a battery icon or a gas gauge. Estimate the percent it shows and explain your benchmark to someone at home.

LESSON 4-3 • HOMEWORK PACKET • SHEET 2 OF 2**Estimate a Percent of a Number**

Maryland Standard 6.RP.A.3c • Learning target – I can estimate a percent of a number using friendly percents and rounded amounts.

REMEMBER

- Round to **friendly numbers** first.
- 10%: divide by 10. 5% = half of 10%.
- 25% = $\frac{1}{4}$, 50% = $\frac{1}{2}$, 75% = $\frac{3}{4}$.

WORKED EXAMPLE

Estimate 24% of 83.

Step 1 Round: 25% of 80.

Step 2 25% = $\frac{1}{4}$, and $80 \div 4 = 20$.

Answer about 20

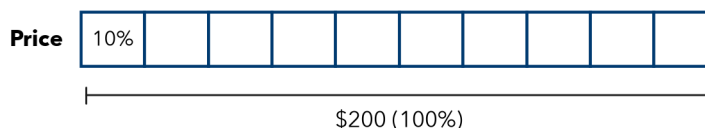
PRACTICE

show your work • label units

1. Estimate. Round to friendly numbers first.

a. 11% of 59 $\approx$ ____ **b.** 74% of 41 $\approx$ ____ **c.** 26% of 203 $\approx$ ____

2. **Sneaker Sale.** Sneakers cost \$198 and are 30% off. Round to \$200. Each part of the tape is 10% of \$200.



- a.** About how many dollars is one part (10%)? _____
- b.** About how much is the discount (30%)? _____
3. **School Lunch.** About 48% of 412 students buy lunch. About how many students is that?
About _____ students
4. **Apple Orchard.** A farm picked 798 apples. About 24% of them were too small to sell. About how many apples were too small?
About _____ apples
5. **Find the Error.** Mia estimated 19% of 512 by writing $20 \times 500 = 10,000$. Explain her mistake. Give a good estimate.
- _____
- _____

TEST PRACTICE

choose the best answer

6. Which is the best estimate of 26% of 79?
A 2 **B** 20 **C** 26 **D** 40
7. A coat costs \$119. It is 40% off. Which is the best estimate of the discount?
A \$4.80 **B** \$40 **C** \$48 **D** \$72

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Find a sale sign. Estimate the discount out loud with someone at home.